

🛒 E-Commerce Analytics | End-to-End Data Engineering Project

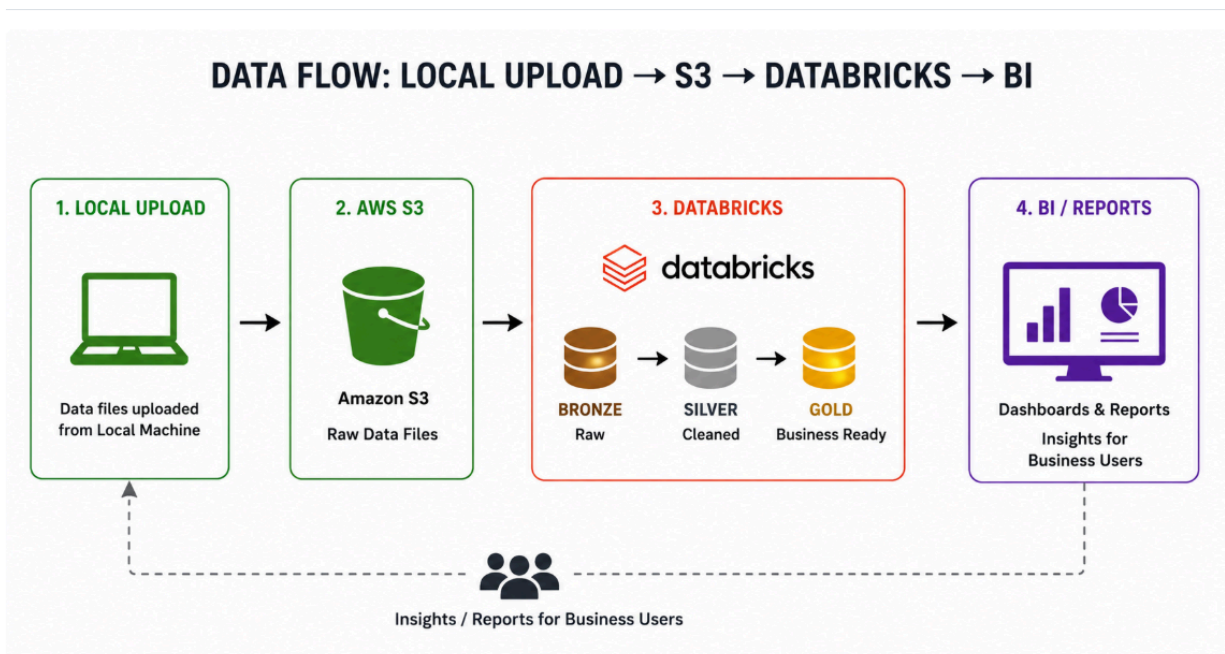
📌 Overview

This project demonstrates an end-to-end **Data Engineering pipeline** built on **Databricks Lakehouse Platform** using the **Medallion Architecture (Bronze → Silver → Gold)**.

The pipeline ingests raw e-commerce transactional data from **AWS S3**, processes it using **Apache Spark**, stores it in **Delta Lake**, and creates business-ready **Dimension** and **Fact** tables for analytics.

The project replaces traditional Python-based ETL scripts with a scalable distributed processing architecture and provides AI-powered analytics through **Databricks Genie** and interactive BI dashboards.

🏗️ Architecture



🚀 Tech Stack

Technology	Purpose
Databricks Free Edition	Lakehouse Platform
Apache Spark	Distributed Data Processing
AWS S3	Data Lake Storage
Delta Lake	ACID Transactions & Time Travel
Unity Catalog	Data Governance
Spark SQL	Analytics
Databricks SQL	Reporting
AI/BI Dashboard	Visualization
Genie AI	Natural Language Query
Databricks Jobs	Pipeline Automation

📁 Project Structure

...

project_ecommerce/

```
├── 1_setup/
│   └── setup_catalog
├── 2_medallion_processing_dim/
│   ├── 2_dim_silver
│   └── 3_dim_gold
├── 3_medallion_processing_fact/
│   └── 3_fact_gold
├── dashboards/
└── README.md
```

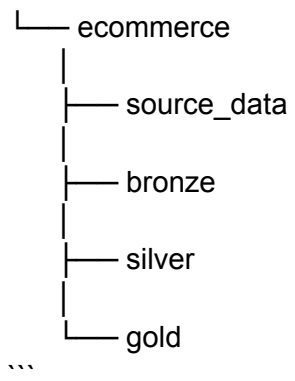
...

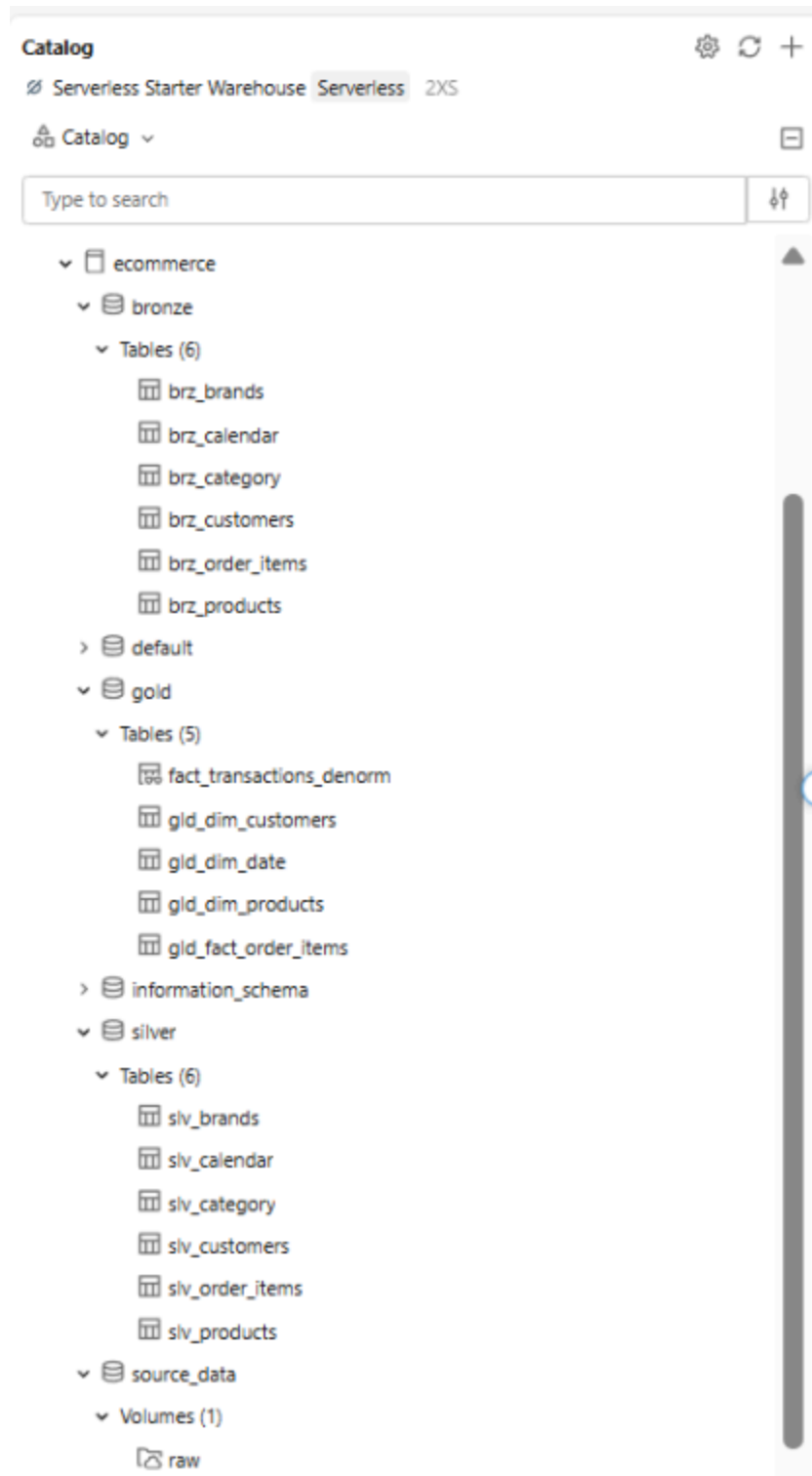
📁 Catalog Structure

...

Catalog

|

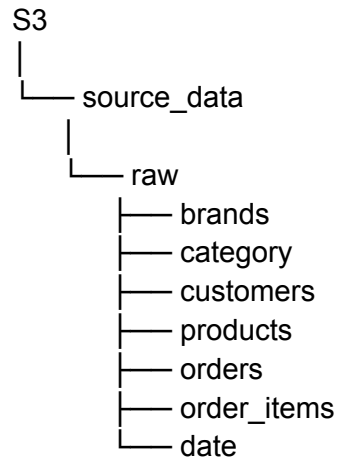




📦 Data Source

Raw CSV files are stored inside AWS S3.

...



...

The screenshot shows the AWS IAM console interface. The top navigation bar includes the AWS logo, a search bar, and the user profile 'usamapatel (957950410306)'. The main content area is titled 'Identity and Access Management (IAM)' and shows the details for a specific role.

Summary

- Creation date:** July 11, 2026, 12:49 (UTC+05:30)
- ARN:** arn:aws:iam::957950410306:role/databricks-uc-7474651308033798-ecommerce112-location
- Last activity:** 10 minutes ago
- Maximum session duration:** 1 hour
- Link to switch roles in console:** <https://signin.aws.amazon.com/switchrole?roleName=databricks-uc-7474651308033798-ecommerce112-location&account=957950410306>

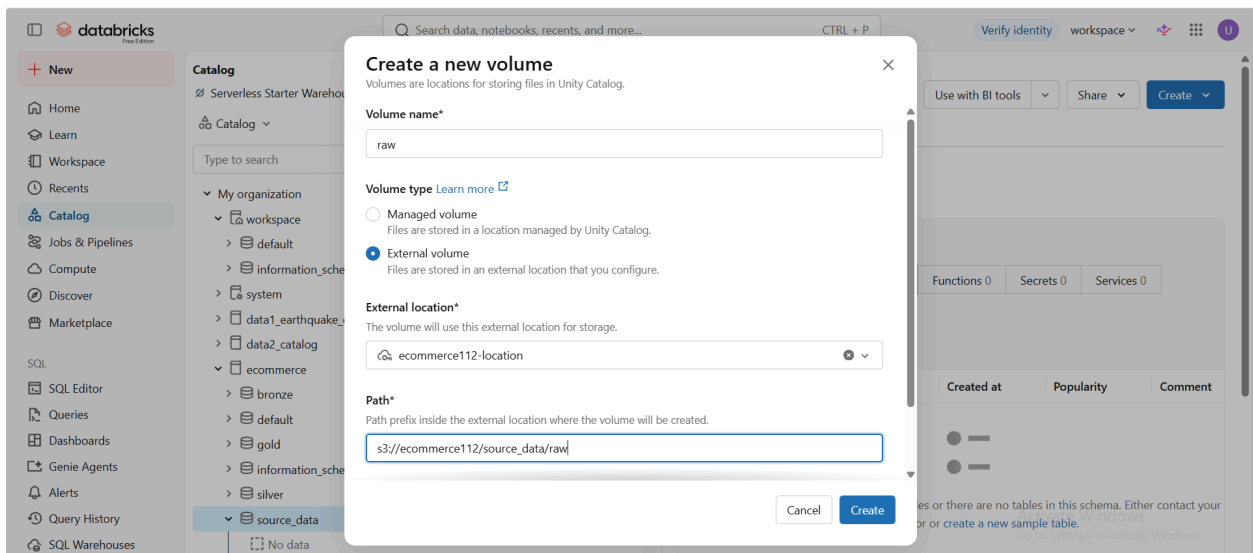
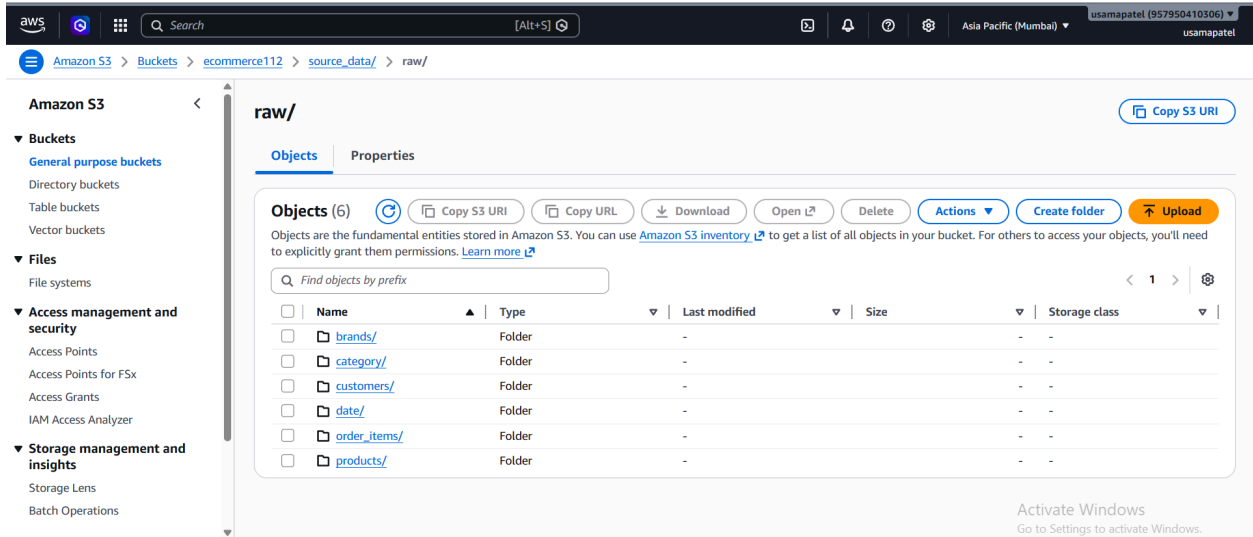
Permissions

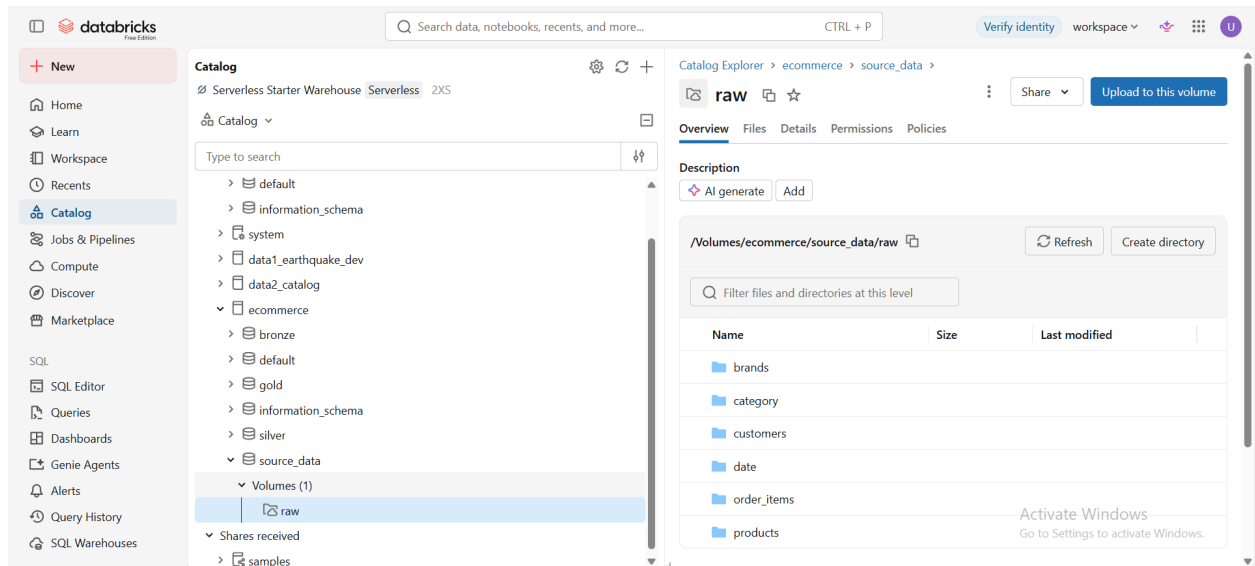
Permissions policies (1) Info

You can attach up to 10 managed policies.

Filter by Type: All types

Policy name	Type	Attached entities
☐ databricks-ext-loc-access-policy-74746513...	Customer inline	0





🏆 Bronze Layer

The Bronze layer stores raw source data without modifying the business data.

Operations

- Raw CSV ingestion
- Schema enforcement
- Delta table creation
- Metadata tracking
- Source file tracking

Metadata Added

- `\_source\_file`
- `\_ingested\_at`

Bronze Tables

- brz_brands
- brz_category
- brz_customers
- brz_products
- brz_orders
- brz_order_items
- brz_date

🏆 Silver Layer

The Silver layer cleans and standardizes the raw data.

Transformations

- Remove duplicate records
- Handle NULL values
- Data type conversion
- Remove special characters
- Standardize text
- Generate business-friendly columns
- Schema validation

Silver Tables

- slv_brands
- slv_category
- slv_customers
- slv_products
- slv_orders
- slv_order_items
- slv_date

🏆 Gold Layer

The Gold layer contains analytics-ready data optimized for reporting.

Dimension Tables

Product Dimension

Contains

- Product ID
- SKU
- Brand

- Category
- Color
- Size
- Material
- Weight
- Rating

Customer Dimension

Contains

- Customer Information
- Demographics
- Country
- Region

Date Dimension

Contains

- Year
- Quarter
- Month
- Week
- Day
- Weekend Flag

Fact Table

fact_transactions_denorm

Denormalized transaction table containing ****34 business columns****.

Transaction Information

- transaction_id
- transaction_date
- transaction_ts

- customer_id
- product_id
- seq_no
- channel

Financial Metrics

- quantity
- gross_amount
- discount_percent
- discount_amount
- tax_amount
- net_amount
- net_amount_inr

Product Attributes

- sku
- category_code
- category_name
- brand_code
- brand_name
- color
- size
- rating_count

Time Dimensions

- year
- quarter
- month_name
- week
- day_name
- hour_of_day
- is_weekend

Promotion Details

- coupon_code
- coupon_flag

Medallion Pipeline

...

AWS S3



External Volume



Bronze



Silver



Gold



Dashboards



Business Users

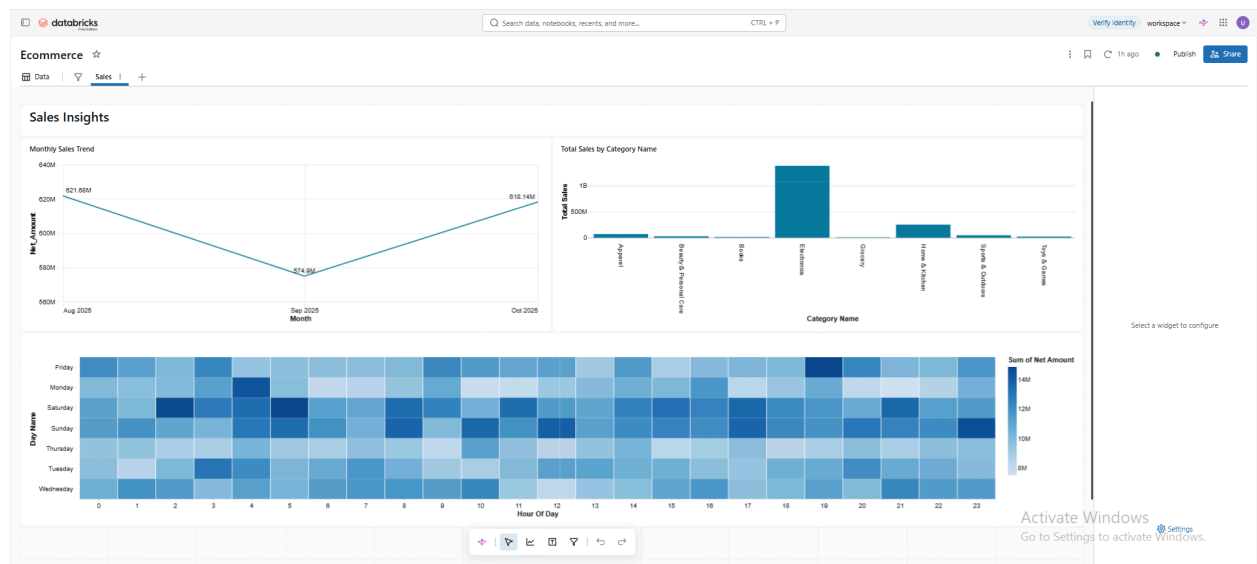
...

Dashboard KPIs

The dashboard provides insights into

- Total Revenue

- Total Sales
- Total Orders
- Monthly Sales Trend
- Revenue by Category
- Revenue by Brand
- Channel Performance
- Coupon Performance
- Customer Purchase Analysis
- Product Performance



📊 Business Metrics

Total Sales

₹1.81 Billion

Top Performing Category

Electronics

₹1.38 Billion

Other Categories

- Home & Kitchen
- Apparel
- Sports & Outdoors
- Beauty & Personal Care
- Toys & Games
- Grocery
- Books

💰 Currency Support

Supported currencies

- INR
- USD

Automatic currency conversion to INR is implemented in the Gold layer.

📱 Sales Channels

- Website
- Mobile

🤖 Genie AI

The Gold layer is connected with ****Databricks Genie**** allowing business users to ask natural language questions.

Example

- Show monthly sales.
- Which category generated highest revenue?
- Top 10 selling products.
- Best customers by revenue.
- Sales trend for Electronics.

Data Governance

Implemented using Unity Catalog.

Features

- Catalog Management
- Schemas
- External Volumes
- Managed Tables
- Data Lineage
- Fine-grained Access Control

Orchestration

Pipeline execution is automated using ****Databricks Jobs****.

...

S3 Upload

↓

Bronze

↓

Silver

↓

Gold

↓

Dashboard Refresh

...

🌟 Key Features

- End-to-End Data Engineering Pipeline
- Medallion Architecture
- Apache Spark Processing
- AWS S3 Integration
- Delta Lake
- Unity Catalog
- Delta Tables
- Dimension & Fact Modeling
- Currency Conversion
- AI-powered Analytics
- Interactive Dashboards
- Databricks Jobs Automation

🎯 Business Use Cases

- Sales Analytics
- Customer Behavior Analysis
- Revenue Analysis
- Product Performance
- Brand Performance
- Category Analysis
- Coupon Effectiveness
- Order Analytics
- Executive Reporting

🌐 Future Enhancements

- Delta Live Tables
- Auto Loader
- Streaming Pipeline
- Customer Cohort Analysis
- Customer Lifetime Value (CLV)
- Product Recommendation Engine
- Inventory Analytics
- ML Demand Forecasting
- CI/CD with GitHub Actions
- Terraform Deployment

 Author

****Usama Patel****

 Email: usamapatel340@gmail.com

****Built using Databricks Lakehouse Platform, Apache Spark, AWS S3, Delta Lake and Unity Catalog.****